Do ratings predict engagement among popular movies?

Investigating whether audience ratings are associated with audience engagement, commercial success, and industry recognition among popular movies listed on TMDb.

Team Lambda

Rita Preto

Tina Bamba

Michael Michie

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Business Case & Hypothesis

Business Case

Streaming platforms and content distributors need to identify movies that are likely to generate audience engagement and deliver strong overall performance. This project investigates whether audience ratings can serve as a useful indicator of audience impact.

Business Hypothesis

Among popular movies listed on TMDb, higher audience ratings are positively associated with:

- Higher audience engagement (IMDb and TMDb votes)
- Greater commercial success (Revenue and Box Office)
- Greater industry recognition (Awards)

QUESTION:

"To what extent do audience ratings predict engagement, commercial success, and industry recognition among movies that have already achieved a certain level of popularity?"

Data Sources and Retrieval Methods

Movies were retrieved from TMDb using the `popularity.desc` sorting option. Therefore, the dataset primarily represents movies with higher visibility and audience exposure rather than a random sample of all films.

TMDb API

Primary dataset

Data Collected

- Movie title
- Release year
- Genre
- TMDb rating
- TMDb vote count
- Popularity score
- Revenue
- Budget
- Runtime

Why We Used It

- Large-scale movie retrieval

OMDb API

Ratings & Engagement

Data Collected

- IMDb Rating
- IMDb Votes
- Box Office Revenue

These variables allow us to compare how audience approval relates to both participation and financial success.

Why We Used It

- Added audience engagement metrics
- Added commercial performance data

Wikipedia Web Scraping

Awards

Data Collected

- Awards Won

Why We Used It

- Added industry recognition metrics
- Complemented ratings and revenue data

By combining APIs and web scraping, the project creates a more complete view of movie performance than any single source could provide.

API and Scraping code

API Code

```python
def fetch_tmdb_movie_details(movie_id, api_key):
    url = f"https://api.themoviedb.org/3/movie/{movie_id}"

    params = {
        "api_key": api_key
    }

    response = requests.get(url, params=params)

    if response.status_code != 200:
        return {}

    return response.json()
```

```python
def fetch_omdb_data(imdb_id, omdb_api_key):

    if not imdb_id:
        return {}

    url = "http://www.omdbapi.com/"

    params = {
        "apikey": omdb_api_key
```

Web scraping code

```python
headers = {"User-Agent": "Mozilla/5.0"}

def check_wikipedia_page(title, year):
    search_url = "https://en.wikipedia.org/w/index.php"

    params = {
        "search": f"{title} {year} film"
    }

    response = requests.get(search_url, params=params, headers=headers)
    soup = BeautifulSoup(response.text, "html.parser")

    page_title = soup.title.text if soup.title else np.nan
    final_url = response.url

    # If Wikipedia returns a search results page, we consider no reliable page found
    if "Search results" in page_title:
        return {
```

Data Preparation: Challenges & Solutions

Initial Dataset: 1000 movies

WRANGLING OBSTACLES AND APPLIED SOLUTIONS

Handling Missing Values

Missing values were identified and handled according to the requirements of each analysis

Data Type Conversion

Numeric variables retrieved as text were converted to appropriate data types

Genre Processing

Genre IDs were mapped to readable genre names and a Primary Genre variable was created

Web Scraping Data Structure

Unlike APIs, Wikipedia pages do not provide data in a standardized format, so relevant award information was extracted from the page content and unnecessary text and formatting were removed.

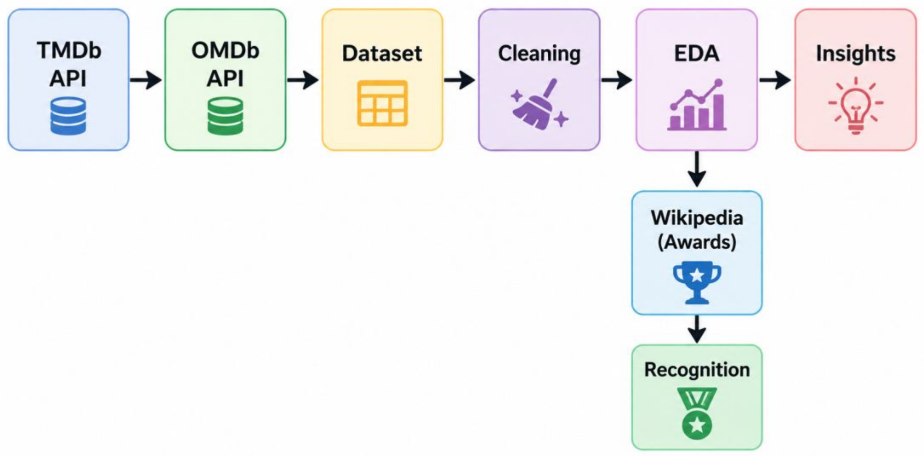
~815 **Movies Retained**
After cleaning and validation

CHALLENGES & LESSONS LEARNED

OMDb alone was insufficient for large-scale movie retrieval, so we combined it with TMDb, using TMDb for movie discovery and OMDb for IMDb ratings, votes, and box-office data.

IMDb review scraping was replaced with awards data due to limited availability and consistency.

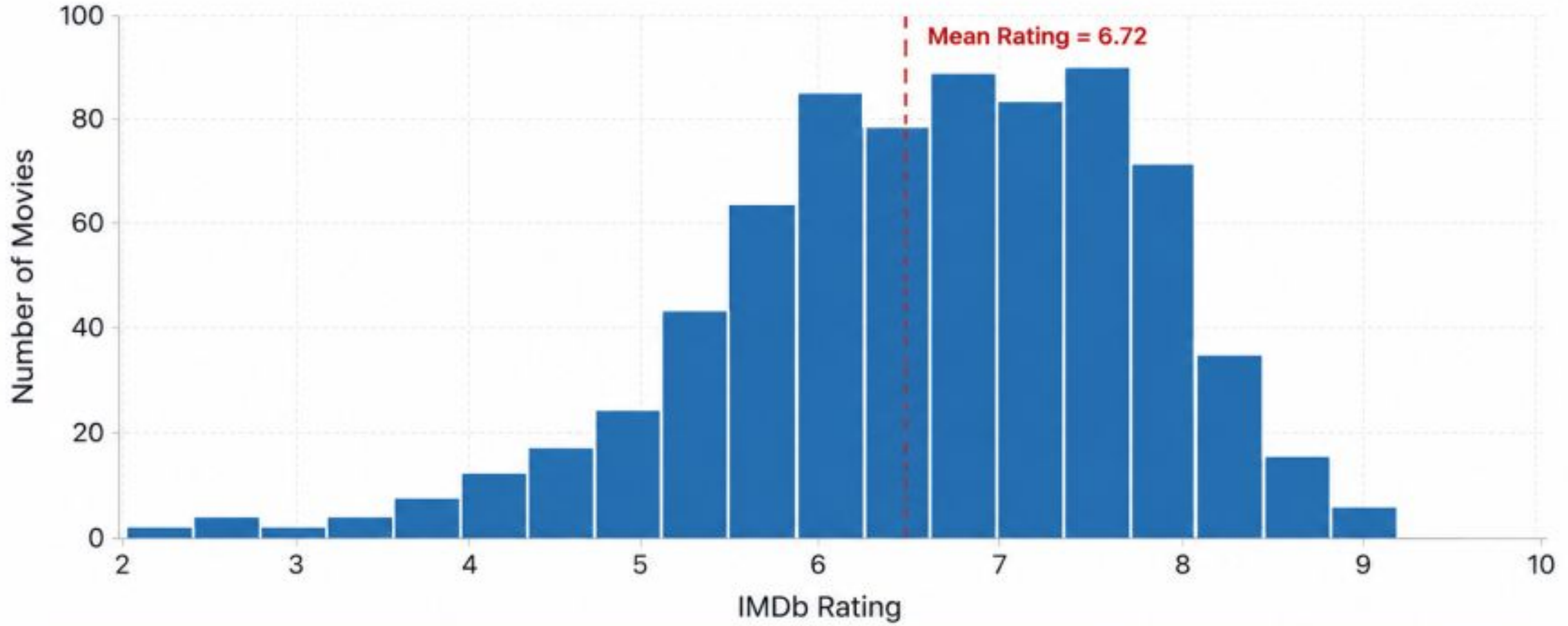
The process



EDA – Rating Distribution

Distribution of IMDb Ratings Across Movies

Most movies are rated between 6 and 8, with a peak around 7.



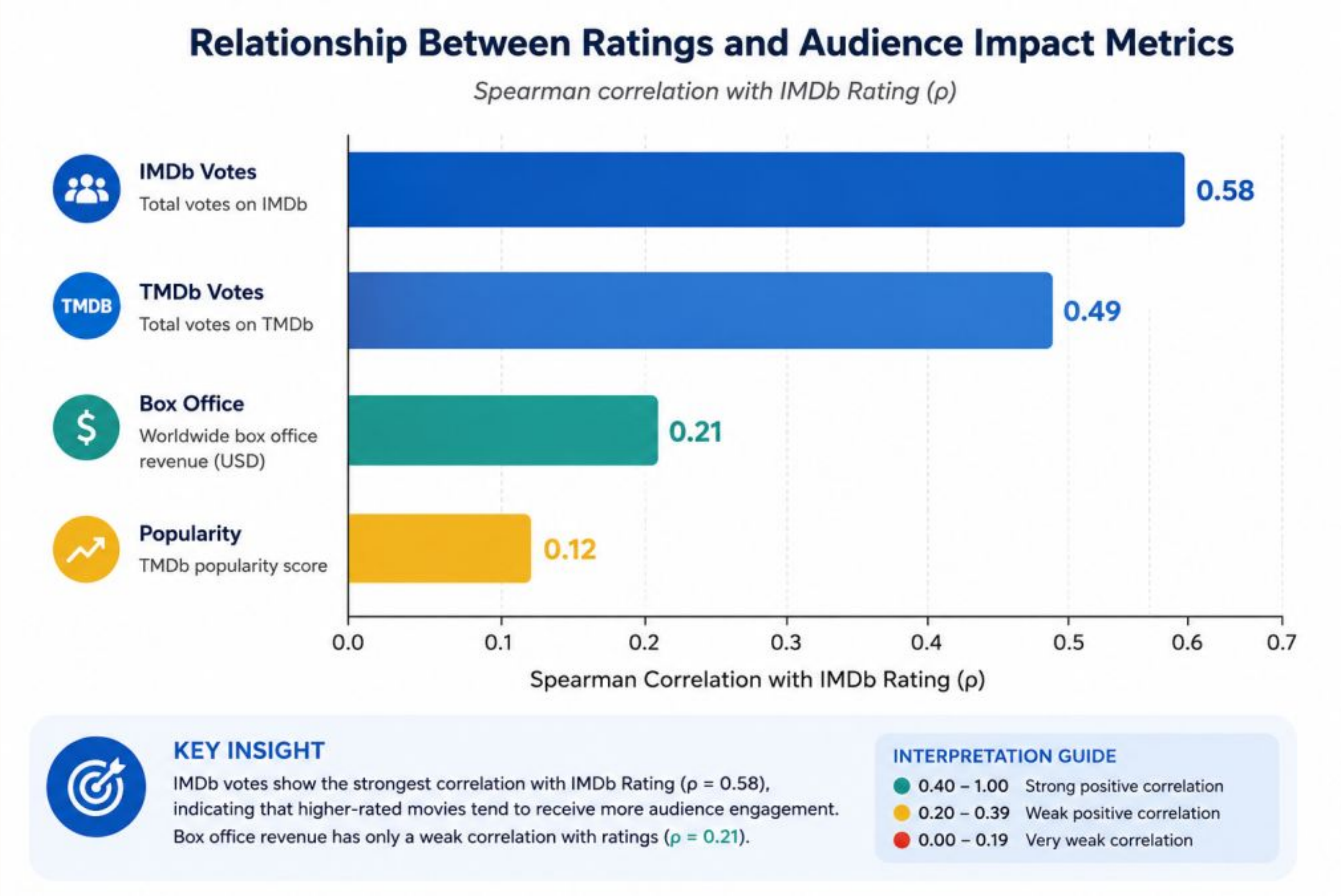
Mean Rating	Median Rating	Std. Deviation	Total Movies
6.72	6.80	1.04	959

Key Insight: Ratings are concentrated between 6 and 8, with few movies receiving extremely low (<3) or very high (>9) ratings.

STATISTICAL METHODS USED

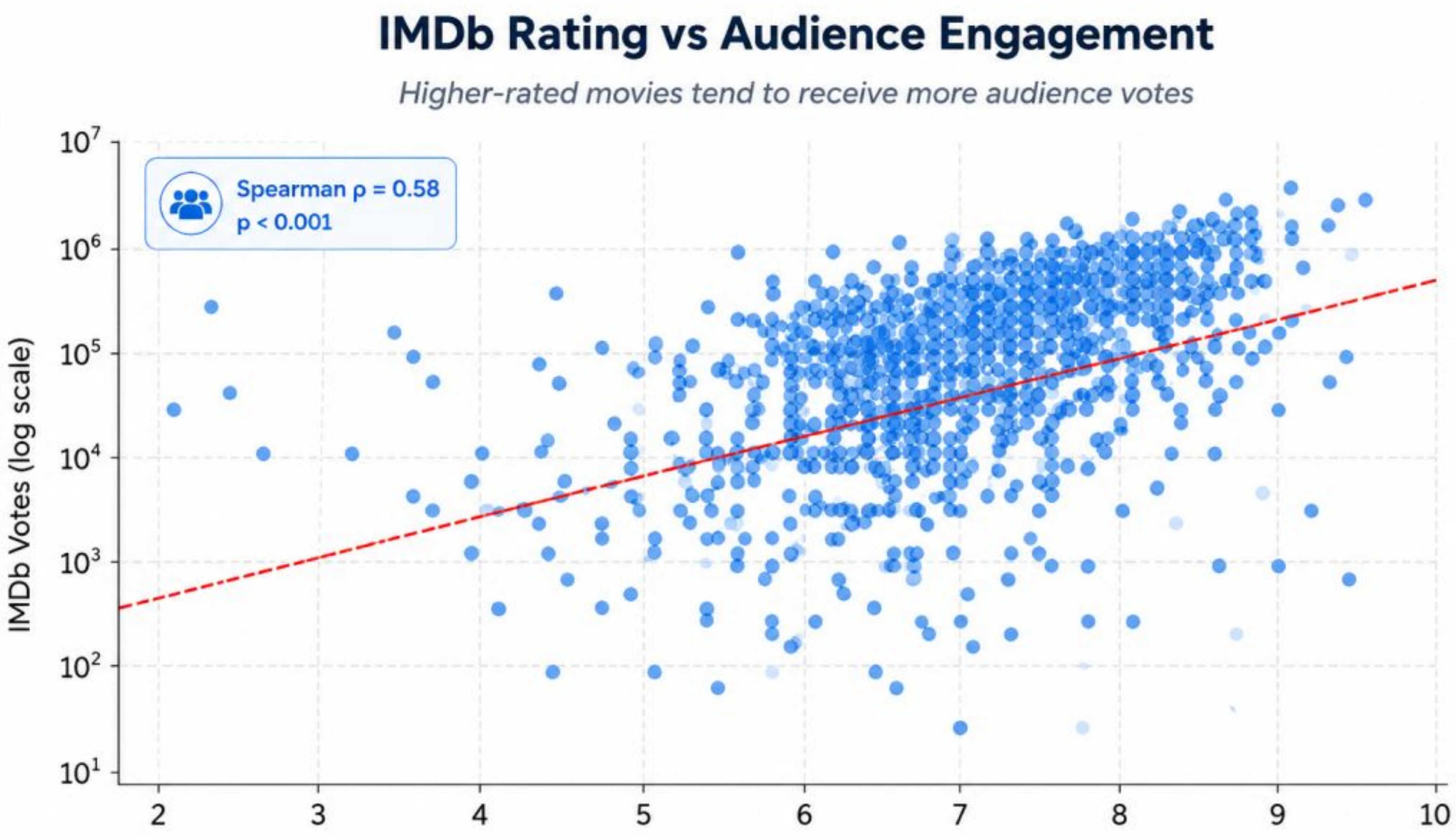
- Descriptive Statistics
- Scatter Analysis
- Spearman Correlation
- Distribution Histograms
- Log Transformation

EDA – Correlation Analysis



Business Results

Ratings Predict Audience Engagement



Business Implication

Higher-rated movies generate substantially greater audience engagement

- Higher-rated movies attract substantially more audience participation.
- Ratings are a strong indicator of audience interest and engagement potential.
- Platforms can leverage ratings to prioritize content for promotion and recommendation strategies.

KEY TAKEAWAY

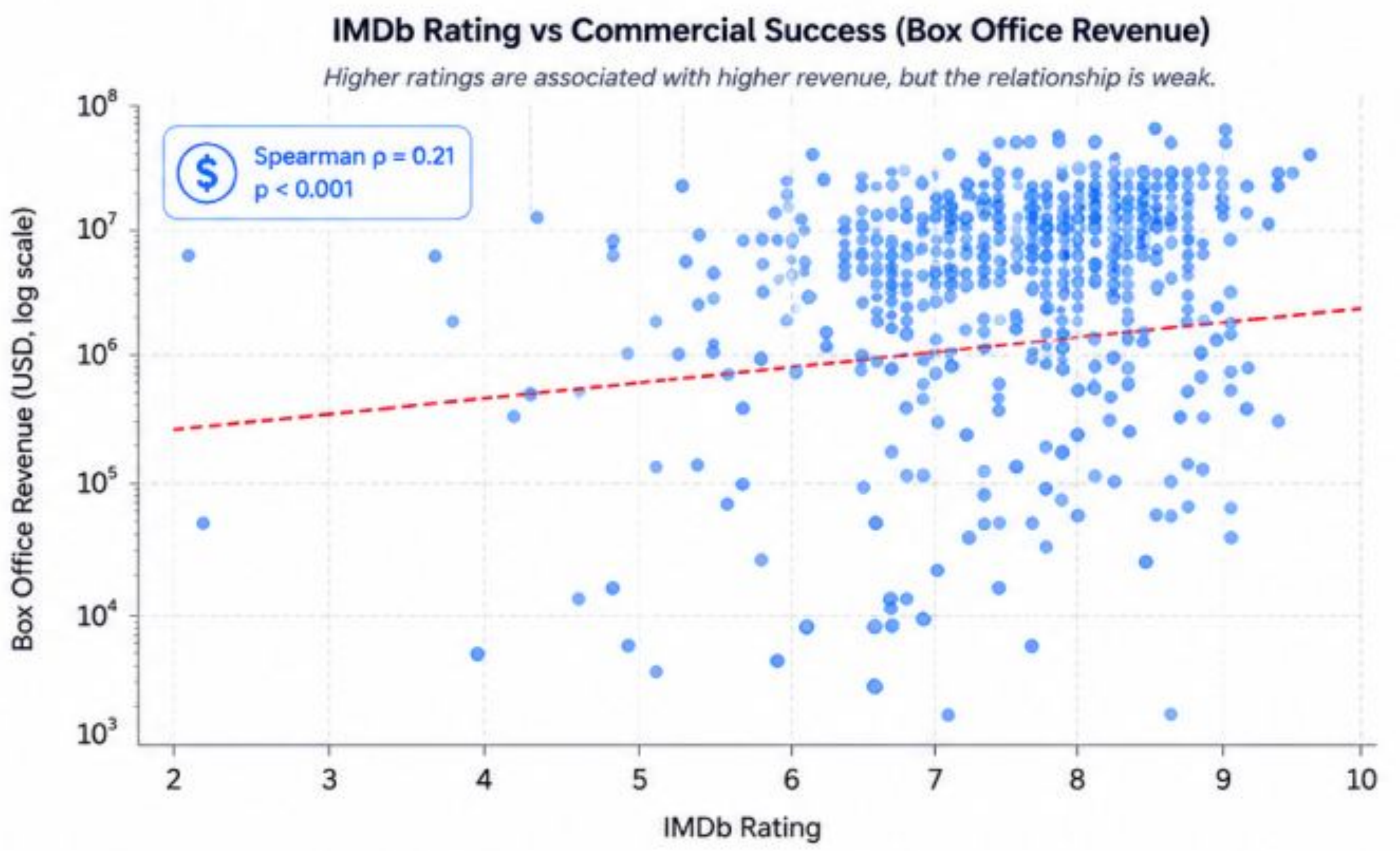
Movies rated above 7 receive significantly more audience votes.
Audience ratings are a meaningful signal for engagement and can support data-driven content decisions.

4.3x
more median votes for movies rated above 7 vs. below 5.5

Note: IMDb Votes on log scale to better visualize distribution across a wide range of values.

Business Results

Ratings Are Not Strong Predictors of Revenue



Business Implication

Higher ratings are associated with higher revenue, but the relationship is weak

- Revenue is likely influenced by additional factors such as marketing, franchise strength, release timing, and production budget
- Many highly rated movies achieve only moderate revenue.
- Some lower-rated movies still generate strong box office results.

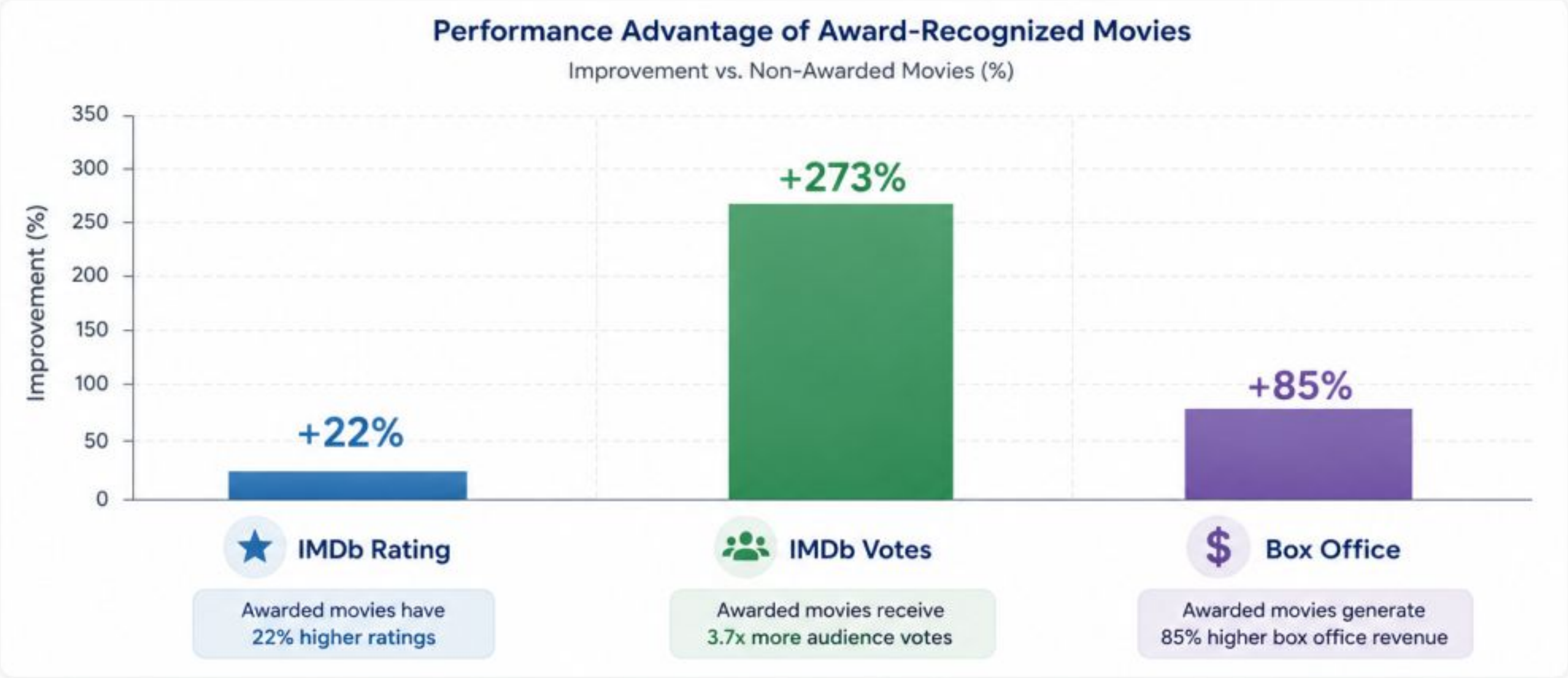
KEY INSIGHT

Ratings alone should not be used to forecast financial performance.
While quality contributes to success, commercial outcomes depend on multiple business factors beyond audience perception.

Marketing Investment | Franchise Strength | Release Timing | Production Budget

Business Results

Award-recognized movies outperform non-awarded movies



This analysis is based on a refined sample of movies released before 2024, selected from their high IMDb vote counts. This ensures more complete public information and awards-related data.

Key Takeaway

Industry recognition is strongly associated with **higher audience perception, greater engagement, and improved commercial success.**

+22%	3.7x	+85%
Higher Ratings	More Votes	Higher Box Office

CONCLUSIONS & STRATEGIC INSIGHTS

Hypothesis Partially Supported: Ratings strongly predict engagement, but are weaker predictors of commercial success.

1 Ratings Drive Engagement

Higher-rated movies receive significantly more audience votes ($\rho = 0.58$)

Ratings can be used as a reliable engagement indicator.

2 Ratings Do Not Guarantee Revenue

Ratings show only a weak relationship with box office performance ($\rho = 0.21$).

Financial success depends on additional factors beyond quality.

3 Awards Signal Strong Performance

Award-recognized movies achieved higher ratings, engagement, and revenue.

Awards can help identify high-value content.

FUTURE RESEARCH & IMPLEMENTATION

- Engagement Prediction
- Awards Analysis
- Review Sentiment Analysis
- Re-watch Metrics

PROJECT CONCLUSION AND Q&A

"Among popular movies, higher-rated movies tend to generate more engagement."

PROJECT TEAM

Team Lambda

- Rita Preto
- Tina Bamba
- Michael Michie

QUESTIONS & DISCUSSION

The floor is now open for questions regarding our methodology, the cinematic data insights, or the future of engagement modeling.